Sampling noisy compressed linear programs

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This construction samples the full endpoint of a deterministic linear program with independent compound-Poisson polynomial noise, even when the program is given by a short grammar with an enormous expansion. The key is to use sparse parity summaries only for counting, while drawing every actual noise mark from the supplied model.

Status. This is a mathematical proof candidate. It has not been formally verified, and neither novelty nor practical speed is established.

1 Model and theorem

Let $V = F_2^D$. The input is an acyclic concatenation grammar whose leaves are either a supplied linear map $C : V \to V$, or additive noise formed by XORing N independent copies of F(U), where N has Poisson mean λ, U is uniform on b supplied bits, and F is an explicit multilinear vector polynomial of degree at most d. All noise occurrences use fresh Poisson and latent randomness, including occurrences of a reused node.

The means λ are nonnegative binary rationals. Binary-encoded repetitions are compiled by repeated squaring and concatenation using O(log N) additional nodes for a repetition count N. Matrix dimensions, explicit monomial lists, latent inputs, and all integer bit lengths are charged to the input. Write G for the compiled grammar size and $H \geq 1$ for its maximum depth.

Theorem. For every $0 < \varepsilon < 1$, the construction samples the full final state to total variation error at most ε in time polynomial in the input length, D, 2^d, and ε^{-1}. For fixed d, the cost is independent of the expanded control length and numerical Poisson means, apart from their bit lengths. The clean output is computed by matrix composition and added to the sampled error frame. The maps need not be invertible.

Final frame intensity

For a node A, let C_A be its clean map and ν_A its mark intensity measure in A's output coordinates. A noise leaf has $\nu_A = \lambda\,\mathrm{Law}(F(U))$; a control leaf has zero intensity. If A is followed by B, independence and linear pushforward give

$$C_{AB} = C_B C_A, \qquad \nu_{AB} = (C_B)_* \nu_A + \nu_B.$$

Thus the final error has compound-Poisson intensity ν_{root}. Marks equal to zero can be discarded throughout. No expanded fault history is required.

2 A parity sparsifier for explicit lists

For a finite positive measure ν and a linear functional ℓ, define the parity cut mass

$$c_\nu(\ell) = \nu\{v : \ell(v) = 1\}.$$

A finite positive weighted-vector summary $\tilde{\nu}$ is ρ-accurate if, simultaneously for every ℓ in V*,

$$e^{-\rho} c_\nu(\ell) \le c_{\tilde{\nu}}(\ell) \le e^{\rho} c_\nu(\ell).$$

Addition of measures and pushforward by any linear map preserve this same simultaneous bound.

Greedy basis peeling

Given nonzero vectors v_i with positive rational weights w_i, i = 1, …, N, repeatedly extract a descending-weight greedy linear basis of the remaining vectors and remove it. Let k_i be the round containing v_i. Each round has at most D vectors. For any cut containing v_i, with total weight C_ℓ,

$$C_\ell = \sum_{j\,:\,\ell(v_j)=1} w_j \;\ge\; k_i w_i.$$

Proof. In each earlier round the greedy basis spans v_i using selected vectors of weight at least w_i. At least one has ℓ-value one. These $k_i - 1$ disjoint contributions, together with v_i itself, prove the bound.

Choose α ≥ 1, retain each vector independently with probability $p_i = \min(1, \alpha/k_i)$, and weight each retained vector by w_i/p_i. For a fixed nonzero cut, every genuinely random summand is bounded by C_ℓ/α and total variance is at most C_ℓ^2/α. Bernstein’s inequality gives, for $0 < \eta \le 1$,

$$\Pr\{|\widetilde{C}_\ell - C_\ell| > \eta C_\ell\} \le 2e^{-\eta^2\alpha/3}.$$

Zero cuts remain exactly zero. A union bound over at most 2^D functionals gives simultaneous relative $1 \pm \eta$ accuracy with failure probability at most δ when

$$\alpha \ge \frac{3(D\log 2 + \log(2/\delta))}{\eta^2}, \qquad \mathbb{E}|S| \le D\alpha(1 + \log N).$$

A fixed polynomial size for merge reduce

Take a loose polynomial bound B, for example $64(D\alpha + 1)^2$, enlarged to hold leaf summaries. A binary merge has at most 2B vectors. Repeat retention until at most B remain. The expected-size bound makes success probability at least one half; conditioning on size success increases the cut-failure probability by at most two. Cap retries logarithmically and charge cap failure to total variation. All work uses rational arithmetic and Gaussian elimination over F_2; no weight-ratio bound is needed.

3 Reusable counts outside every subspace

Polynomial noise leaves

A nonzero scalar multilinear polynomial of degree at most d on uniform Boolean inputs equals one with probability at least $2^{-\mathrm{d}}$. To see this, choose a monomial of maximum degree r. After assigning all other variables, that top monomial survives, so the restricted polynomial is nonzero on at least one of its 2^{r} inputs.

Therefore $\ell(\mathrm{F}(\mathrm{U}))$ is either identically zero or has one-probability at least $2^{-\mathrm{d}}$. Multiplicative Chernoff bounds and a union bound over the 2^{D} functionals imply that

$$M = O\left(2^{d}\frac{D + \log(1/\delta)}{\eta^{2}}\right)$$

fresh uniform latent samples, with weight λ/M on each image, approximate all leaf cuts to relative 1 ± η with failure probability at most δ. The exact coefficient span can be computed separately for deterministic zero tests.

Bottom up compilation

At a concatenation, push the left summary through the clean right matrix, unite it with the right summary, and sparsify the resulting explicit list. Store one summary for each reused node rather than expanding its occurrences. Allocate a small failure budget to each leaf and compression.

Conditional on success of all summaries, errors multiply along a path. If η ≤ 1/2, every node has simultaneous cut error bounded by

$$\rho = 2H\eta.$$

A path has at most H local factors 1 ± η; addition and pushforward add no further multiplicative loss. Fresh randomness at each compression allows conditional failure bounds and a union bound over G nodes. There is no need to enumerate suffix contexts: every summary already controls every functional in its D-dimensional space.

The outside subspace identity

For any subspace W ≤ V, let $\mathrm{W}^{\perp}$ denote its annihilator in V*. Then

$$\nu(V \setminus W) = 2\,\mathbb{E}_{\ell\ \text{uniform on}\ W^{\perp}} c_{\nu}(\ell).$$

Proof. If v lies in W, every such ℓ vanishes on v. If v lies outside W, exactly half of these functionals have $\ell(\mathrm{v}) = 1$. Average first for each v, then sum its mass.

Consequently a ρ-accurate parity summary gives the identical relative accuracy for ν(V \ W). It also estimates $\nu\{\mathrm{v} : \mathrm{Lv} \notin \mathrm{W}\}$, for every linear context L: compose each functional with L, or take the preimage of W. Compute this query by summing summary weights whose transformed vectors fall outside W.

Adaptive guarantee. Both W and L may depend on previous marks and the summaries themselves. The simultaneous cut guarantee already covers these choices; no further union bound over subspaces or contexts is required.

4 True conditional marks and rank growth

Descent through the original grammar

To acquire a true root mark conditioned on lying outside W, descend the actual grammar. Maintain a context matrix L from the current node's output to root coordinates. At a concatenation A followed by B, the two active masses are

$$a_A = \nu_A\{v : LC_B v \notin W\}, \qquad a_B = \nu_B\{v : Lv \notin W\}.$$

Query the two summaries and select a branch in proportion to the estimated active masses. On the left update L to LC_B; on the right keep L. On successful preprocessing, true zero masses are estimated as zero. Exact span tests can also guard zero cases.

If each estimate is within $e^{\pm\rho}$, normalizing changes branch probabilities by $O(\rho)$ in total variation. There are at most H branches. Hence the selected leaf and context differ from the true active conditional selection by $O(H\rho)$.

At a selected polynomial noise leaf, draw **fresh** uniform latent inputs until LF(U) lies outside W. These inputs are independent of preprocessing samples. If acceptance is possible, some functional annihilating W makes $\ell(\mathrm{LF}(U))$ a nonzero degree-at-most-d polynomial. Acceptance probability is therefore at least 2^{-d}. Rejection sampling gives the exact conditional law at that leaf in at most 2^d expected proposals. A cap of $O(2^d \log(D/\varepsilon))$ proposals adds a chosen small total failure probability.

Distribution preservation. Only the branch selection is approximate. Given the selected leaf and context, the acquired mark has its true conditional distribution. The sparse summary is never substituted for the physical jump law.

A finite rank representation

For a jump by mark v of intensity $\nu(v)$, replace it with a rate-$2\nu(v)$ proposal that adds either zero or v, each with probability one half. The generator is unchanged:

$$2\nu(v)\left[\frac{f(x)+f(x+v)}{2} - f(x)\right] = \nu(v)[f(x+v) - f(x)].$$

Conditional on the proposed marks, the accumulated error is uniform on their linear span W: it is the image of independent fair bits under a linear map. A proposal inside W leaves this conditional law unchanged and can be discarded. A proposal outside W enlarges the span by one dimension.

Thus at most D active stochastic transitions are needed to describe the final error law. This is a law-level representation of the endpoint; it does not enumerate, reconstruct, or preserve the original fault history.

5 The endpoint sampler and its error

The exact auxiliary sampler

Run the following process for a time horizon of one, with $\nu = \nu_{\mathrm{root}}$.

1. Start with $W = \{0\}$ and elapsed time $t = 0$.
2. Set $a = 2\nu(V \setminus W)$. If $a = 0$, stop.
3. Draw an exponential waiting time T with rate a. If $T > 1 - t$, stop; otherwise set $t \leftarrow t + T$.
4. Draw a true mark conditional on being outside W, and adjoin it to W.
5. Repeat. On stopping, output a uniform vector of W.

The previous page proves this produces the exact compound-Poisson endpoint. Uniform sampling from W means choosing fair coefficients for any basis. Add the clean deterministic output to obtain the full final state. Each accepted transition increases dimension, so there are at most D transitions and at most D clock draws.

Approximate rates and conditional acquisition

Replace the rate query by the root summary and the conditional-mark step by grammar descent. Two exponential laws whose rates differ by $e^{\pm\rho}$ have $O(\rho)$ total variation distance, uniformly in the numerical size of the rate. Censoring at the remaining horizon cannot increase this distance. The conditional acquired mark has error $O(H\rho)$.

A sequential coupling, applied while both histories agree and using the adaptive query guarantee, gives

$$d_{\mathrm{TV}} \le O(DH\rho) + \delta_{\mathrm{prep}} + \delta_{\mathrm{caps}} + \delta_{\mathrm{arith}}.$$

Here preprocessing failure includes all leaf and compression failures; cap failure includes bounded retries and latent rejection; arithmetic error covers finite-bit random choices and clocks. Choose a sufficiently small absolute constant c and set

$$\eta = \frac{c\varepsilon}{DH^2}, \qquad \rho \le 2H\eta.$$

Distribute the remaining error budget among preprocessing, caps, and arithmetic. The resulting total variation error is at most ε. For example, $\eta = \varepsilon/(100DH^2)$ can be reduced by a harmless absolute factor to absorb the constants in the coupling bound.

Why parity preservation is not enough on its own

This argument makes no spectral-to-endpoint-total-variation inference. Preserving all Fourier exponents by a small generator sparsifier does not, by itself, preserve the full finite-time law in total variation. Here the summary estimates only clock rates and branch masses. Every sampled mark comes from the original input model, which is the essential extra step.

On any explicitly bounded failure event, the implementation may return a fixed vector; its probability is charged directly to the error budget. All success guarantees remain statements about the unconditional output distribution.

6 Bit complexity and scope

Rational data and finite clocks

All summaries have polynomially many vectors. Basis peeling adds rational factors k/α with polynomial bit descriptions; numerator and denominator lengths grow only polynomially across H levels. Expanded occurrence counts are never stored in unary. Matrices, spans, coefficient evaluation, outside-subspace tests, and branch weights use exact finite-field or rational arithmetic. Categorical choices can use integer rejection sampling or a bounded approximation whose error is budgeted.

Real-valued clock primitives are unnecessary. Let A be twice the total positive weight in the root summary. Every approximate active rate is at most A, and every nonzero rate is at least twice the smallest positive stored weight. Both extremes have polynomial bit descriptions. The zero-summary case stops immediately.

Generate each clock to absolute error Δ, treating overflow beyond two as a stop. With at most D clocks, rounding can change the stopping index only if a cumulative arrival is within DΔ of time one. Conditional on its preceding history, an arrival has density at most A. A union bound gives probability $O(D^2A\Delta)$; choose

$$\Delta = O\left(\frac{\varepsilon}{D^2 \max(1, A)}\right).$$

For $T = -\log(U)/a$, exclude a tiny-U event of probability $O(\varepsilon/D)$, truncate uniform bits, and evaluate the logarithm to the needed precision using the known positive lower rate bound. The precision is polynomial in log A, the logarithm of the smallest positive stored rate, and $\log(1/\varepsilon)$. It never scales with A itself. This accounts for binary-encoded huge rates rather than assuming an ideal real-clock oracle.

What the construction supplies

The output is one full joint endpoint error frame. It may be passed through a supplied polynomial-time syndrome or decoder computation, retaining an additive distributional guarantee. Nested repetitions, heterogeneous tiny or huge means, and directions appearing in only an exponentially rare expanded occurrence are included. Clifford schedules are a physical application when their supplied matrices are the binary symplectic actions on Pauli frames.

Assumptions and unresolved claims

The control grammar, explicit polynomial noise maps, and rate calibration must be supplied or acquired separately. The theorem does not cover coherent noise, state-dependent rates, adaptive measurement feedback, or the expanded measurement trajectory. It gives no relative accuracy guarantee for exponentially rare decoder failures. The noise interface is an assumption, not a tomography or physical-validation result.

The proposed combination is uniformly reusable parity-based active-mass counting, conditional descent through the original compressed program, and finite-rank heatbath innovations. The argument does not establish a priority claim. Its polynomial exponents may be large, and no implementation benchmark or practical-speed claim is provided.

Discrete finite-alphabet extensions are being developed separately and are not used in this proof.

Related literature: S. Khanna, A. Putterman, and M. Sudan, *Code Sparsification and its Applications*, arXiv:2311.00788. A comprehensive literature comparison remains open.